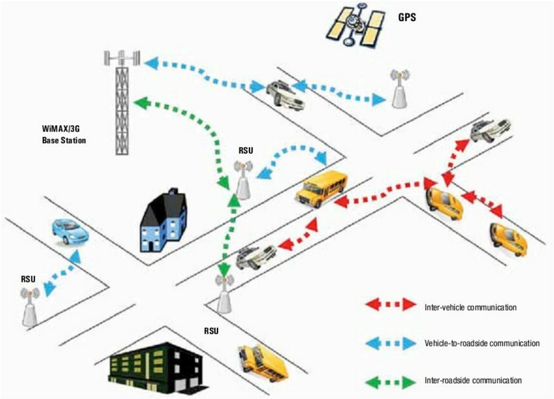

VANET Simulation Using FOSS

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Part—1

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As the name suggests, VANET is a network wherein the nodes are vehicles equipped with on-board units (OBU) for networking. A VANET may also have Road-Side Units (RSU), which are static infrastructure nodes to support the network's functionality. VANET communication comes in two flavours: Vehicle-to-Vehicle (V2V) communication and Vehicle-to-Infrastructure (V2I) communication. A typical VANET scenario is shown in Figure 1 and the VANET architecture can be seen in Figure 2.

Although it is a type of mobile ad-hoc network, VANET exhibits its own distinctive characteristics, such as:

- The nodes move at very high speed.
- It has a highly dynamic topology with very high node density.
- The network is frequently disconnected.
- Mobility patterns are constrained.
- There are hard delay constraints.
- It involves interaction with on-board sensors.

Some of the applications of VANET to avoid collisions include traffic signal violation warnings, stop sign violation warnings, left turn assistance, stop sign movement assistance, intersection collision warnings, blind merge warnings, pedestrian crossing information, etc. There are many such applications, ranging from safety-critical features to infotainment. The research done in this area cannot be tested 'live' on the road, due to the safety implications. We need to set up a simulation environment for such network testing. There are two components of VANET simulation:

- The mobility component: Defining road topology and vehicle movement
- The networking component: Network traffic simulation

There are various VANET simulators available, as shown in Figure 3. This article focuses on the open source Network Simulator-2 (NS2) to simulate the networking component,